

PhishGuard AI

AI-Based Phishing Email & Malicious URL Detection

Artificial Intelligence and Machine Learning - II

Asim Khan · 231407 | Arshan Attar · 231408

HOD: Dr. Zainab Mirza



The Problem

Phishing attacks trick users into revealing sensitive information or clicking malicious links.



Fake login email

“Your account will be suspended. Verify now.”



Suspicious URL

<https://secure-account-verification.example.com/login>



Urgent payment message

“Payment failed. Update your billing information immediately.”

Manual detection is slow and difficult because phishing emails and malicious URLs can closely resemble legitimate ones.

Our Solution and Architecture

PhishGuard AI combines machine learning-based email classification with AI-based malicious URL detection in a single application.

Main flow



Email branch



URL branch



How It Works

Both models analyze input patterns and provide an automated detection result.

Engine 1 · Email phishing detection

#	Step	Detail
1	Data	Multiple email datasets combined
2	Preprocess	Text cleaned and prepared
3	Features	TF-IDF converts text to numerical features
4	Model	Logistic Regression predicts category
5	Output	Phishing / Legitimate

Engine 2 · Malicious URL detection

#	Step	Detail
1	Data	UCI Phishing Websites Dataset
2	Features	30 phishing-related features
3	Preprocess	Numerical conversion and feature scaling
4	Model	Dense network with ReLU and sigmoid output
5	Training	Training for 50 epochs
6	Output	Phishing probability and classification

Results

98.48%

Logistic Regression · Email detection accuracy

82,486 emails combined dataset

Confusion matrix

	Predicted Legitimate	Predicted Phishing
Actual Legitimate	7795	124
Actual Phishing	126	8453

The email classifier achieved strong performance on the held-out test dataset.

URL detection

UCI Phishing Websites Dataset

11,055

samples

30 features

Dense Neural Network
50 epochs

URL model trained for phishing
probability prediction.

No URL accuracy, precision, recall or F1 is
shown.

Innovation, Limitations & Future Scope

Innovation

- Combined email and URL phishing detection
- AI-based automated threat analysis
- Risk percentage for easier interpretation
- Simple user-friendly detection interface
- Scan history and detection explanation

Limitations

- URL detection depends on available extracted features
- Some webpage-level features cannot be reliably obtained from a raw URL alone
- Dataset-based models may behave differently on unseen attacks
- Current system is an academic prototype

Future Scope

- Real-time threat intelligence
- Browser extension
- Email client integration
- Larger real-world phishing datasets
- Advanced transformer-based email detection
- Improved webpage-based URL analysis
- Cloud-based deployment



Thank You

PhishGuard AI · AI-Based Phishing Email & Malicious URL Detection

Made with **GAMMA**